Natural characteristic smooth duality for supersingular representations of $\mathrm{GL}_2(\mathbb{Q}_p)$

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1 Introduction

Let G denote a locally profinite group, E a field, and let $\mathrm{Rep}_E^\infty(G)$ denote the category of smooth G-representations over E. Given an object V of $\mathrm{Rep}_E^\infty(G)$, one can define its *smooth dual* $S^0(V)$ to be the subspace of smooth vectors in the full dual space $\check{V} = \mathrm{Hom}_E(V, E)$, which admits a natural action of G. When the characteristic of E is zero, this is a useful tool for studying smooth representations over E. However, in natural characteristic, this is much less useful; as shown by Kohlhaase in [Koh17], $S^0(V)$ vanishes in many cases when G is a p-adic Lie group and $\operatorname{char} E = p$. In that paper, he develops a family of functors called the *higher smooth duals*, and proves many results about them, in particular computing their values on certain supersingular representations of $\mathrm{GL}_2(\mathbb{Q}_p)$.

In Section 2, we will review some of the necessary background on Pontryagin duality. In Section 3, we will define Kohlhaase's higher smooth duals, and present a specific result showing why the smooth dual S^0 on its own is insufficient. In Section 4, we will provide a sketch of Kohlhaase's computation of the higher smooth duals when applied to certain supersingular representations of $\mathrm{GL}_2(\mathbb{Q}_p)$.

2 Pontryagin duality

In this section, we will review the relevant material from [Koh17, Section 1].

Notation 2.1. In this section, G will denote a locally profinite topological group, and E will be any field. We will write $\mathrm{Rep}_E^\infty(G)$ for the category of smooth G-representations over the field E. Write also TopVec_E for the category of topological E-vector spaces. Given a ring R, we write Mod_R for the category of left R-modules.

From now on, we will endow E with the discrete topology.

Definition 2.2. An E-vector space is *discrete* if it is endowed with the discrete topology. If it is an inverse limit (in the category TopVec_E of topological vector spaces) of finite dimensional discrete E-vector spaces, then it is *pseudocompact*. Let Disc_E and Pct_E denote the full subcategories of TopVec_E whose objects are discrete and pseudocompact, respectively.

An essential fact is that Pct_E is an abelian category (see e.g. [Gab62, Théorème IV.3.3]).

Definition 2.3. If $V \in \mathrm{Disc}_E$, and $(V_i)_{i\in I}$ is its family of finite dimensional subspaces, then write

$$\check{V} \coloneqq \mathrm{Hom}_E^{\mathrm{cts}}(V, E) = \mathrm{Hom}_E(V, E)$$

for its dual, which can be understood as a pseudocompact E-vector space via the isomorphism

$$\check{V} = \mathrm{Hom}_E(V, E) \cong \varprojlim_{i\in I} \mathrm{Hom}_E(V_i, E).$$

Conversely, if $W \in \mathrm{Pct}_E$, we can equip its dual $\check{W} = \mathrm{Hom}_E^{\mathrm{cts}}(W, E)$ with the discrete topology. One then has the following theorem:

Theorem 2.4. *([Koh17, Theorem 1.1]) The functors* $\check{(\cdot)}$ *are mutually quasi-inverse equivalences between the categories* Disc_E *and* Pct_E.

The details of the proof are standard and not especially interesting. We remark that if V is discrete, then choosing a basis gives an isomorphism

$$V \xrightarrow{\sim} \bigoplus_{j \in J} E$$

where the right side is equipped with the discrete topology. Then $\check{V} = \prod_{j \in J} \mathrm{Hom}_E(E, E) = \prod_{j \in J} E$ is endowed with the product topology; one can then check that $\check{\check{V}} \cong V$, as any continuous map from $\prod_{j \in J} E$ to the discrete space E has open kernel, and hence factors through the projection onto the product of only finitely of the copies of E. A similar strategy works for showing the other composition is also naturally isomorphic to the identity.

Definition 2.5. For a profinite group G_0, we define its *Iwasawa algebra* to be

$$\Lambda(G_0) = E[\![G_0]\!] := \varprojlim_{N \trianglelefteq G_0} E[G_0/N],$$

endowed with the pseudocompact topology. More generally, we define the *augmented Iwasawa algebra* of the locally profinite group G to be

$$\Lambda(G) := E[G] \otimes_{E[G_0]} \Lambda(G_0)$$

for $G_0 \leq G$ any open compact subgroup; a priori, this is only a vector space, however Kohlhaase shows on [Koh17, p. 6] that it admits a unique E-algebra structure such that the maps

$$\delta \mapsto \delta \otimes 1 : E[G] \to E[G] \otimes_{E[G_0]} \Lambda(G_0)$$
$$\delta' \mapsto 1 \otimes \delta' : \Lambda(G_0) \to E[G] \otimes_{E[G_0]} \Lambda(G_0)$$

are E-algebra homomorphisms. Moreover, it is independent of the choice of G_0 up to unique isomorphism.

Notation 2.6. We write $\mathrm{Mod}^{\mathrm{pct}}_{\Lambda(G)}$ for the category of pseudocompact E-vector spaces M with a continuous left G-action, i.e., the map $G \times M \to M$ is jointly continuous (with respect to the product topology). This notation is justified by the following standard lemma:

Lemma 2.7. *([Koh17, Lemma 1.3 and Corollary 1.4]) Let $M \in \mathrm{Pct}_E$ be equipped with an action of G by continuous E-linear automorphisms. The following are equivalent.*

1. *$M \in \mathrm{Mod}^{\mathrm{pct}}_{\Lambda(G)}$, i.e., the map $G \times M \to M$ is jointly continuous.*

2. *If $G_0 \leq G$ is a compact open subgroup, then the $E[G_0]$-module structure of M extends to a $\Lambda(G_0)$-module structure for which the map $\Lambda(G_0) \times M \to M$ is jointly continuous and such that M admits a basis of neighborhoods of 0 consisting of $\Lambda(G_0)$-submodules.*

In particular, if $M \in \mathrm{Mod}^{\mathrm{pct}}_{\Lambda(G)}$, then the $E[G]$-module structure of M extends uniquely to a $\Lambda(G)$-module structure; moreover, any morphism in $\mathrm{Mod}^{\mathrm{pct}}_{\Lambda(G)}$ is $\Lambda(G)$-linear.

We then have a crucial theorem:

Theorem 2.8. *([Koh17, Theorem 1.5]) The functors $\check{(\cdot)}$ restrict to quasi-inverse equivalences of categories between* $\mathrm{Rep}_E^\infty(G)$ *and* $\mathrm{Mod}_{\Lambda(G)}^{\mathrm{pct}}$.

Proof. We will briefly sketch the proof. Fix a compact open subgroup $G_0 \leq G$. If $V \in \mathrm{Rep}_E^\infty(G)$, then G acts on $\check{V}$ by continuous E-linear automorphisms. Because V is smooth, it can be written as a direct limit of smooth finite-dimensional G_0-subrepresentations $V = \varinjlim_i V_i$; now

$$\check{V} \cong \varprojlim_i \check{V}_i,$$

and each of the $\check{V}_i$ is a finite-dimensional discrete topological G_0-module. One can then invoke Lemma 2.7 to conclude that $\check{V} \in \mathrm{Mod}_{\Lambda(G)}^{\mathrm{pct}}$. The other direction is similar. □

We now define the 0-*th smooth duality functor*; the notation will be clarified later when we work with higher smooth duals.

Definition 2.9. If M is any $E[G]$-module and if $H \leq G$ runs over the open subgroups of G, then we define

$$\Sigma^0(M) = \Sigma^0_G(M) \coloneqq \varinjlim_H M^H \in \mathrm{Rep}_E^\infty(G)$$

to be the G-subrepresentation of M consisting of all smooth vectors. We then define the 0-*th smooth dual* of a smooth representation $V \in \mathrm{Rep}_E^\infty(G)$ to be

$$S^0(V) = S^0_G(V) \coloneqq \Sigma^0_G\left(\check{V}\right),$$

which yields a functor $S^0 : \mathrm{Rep}_E^\infty(G) \to \mathrm{Rep}_E^\infty(G)$.

3 Higher smooth duals

We now assume that we are working in *natural characteristic*, i.e., that $\mathrm{char}\, E = p \neq 0$ and G is a p-adic Lie group.

Unfortunately, the 0-th smooth dual yields little information in this setting; this is in stark contrast to the non-natural characteristic situation (see [Koh17, Section 2]). To make this more precise, let $\mathrm{Rep}_E^\infty(G)^a$ denote the full subcategory of $\mathrm{Rep}_E^\infty(G)$ consisting of *admissible* smooth representations V, i.e., those V satisfying $\dim_E V^H < \infty$ for any open subgroup $H \leq G$.

Lemma 3.1. *Fix any compact open $G_0 \leq G$, and say $V \in \mathrm{Rep}_E^\infty(G)^a$. Then $\check{V}$ is a finitely generated $\Lambda(G_0)$-module.*

Proof. (This is one part of the proof of [Koh17, Proposition 1.9].) Shrinking G_0 merely makes the result stronger, so without loss of generality, suppose that G_0 is pro-p. Then $\Lambda(G_0)$ is a local ring with maximal ideal $I_{G_0} = \ker(\Lambda(G_0) \to E)$ (e.g., see [NSW08, Proposition 5.2.16(iii)]). By the topological Nakayama lemma ([Bru66, Corollary 1.5]), it is now sufficient to show that $I_{G_0}\check{V}$ has finite codimension in $\check{V}$, i.e., that

$$\check{V}/I_{G_0}\check{V} \cong E \otimes_{\Lambda(G_0)} \check{V}$$

is finite dimensional. Notably, one can easily verify that

$$E \otimes_{\Lambda(G_0)} \check{V} \cong (V^{G_0})\check{},$$

and the latter is certainly finite dimensional by admissibility, giving the claim. □

We note that the proof of the above lemma only requires G to be locally pro-p, not necessarily a p-adic Lie group as we are assuming in this section.

Proposition 3.2. *([Koh17, Proposition 3.9]) If $V \in \mathrm{Rep}_E^\infty(G)^a$, then $S^0(V)$ is finite dimensional over E. In particular, if V is irreducible and infinite dimensional over E, then $S^0(V) = 0$. The largest full subcategory of $\mathrm{Rep}_E^\infty(G)^a$ on which the 0-th smooth dual S^0 restricts to an anti-equivalence of categories is the category of finite dimensional smooth representations of G over E.*

Proof. This proof is given in [Koh17, Remark 3.10]. Fix an open pro-p subgroup $G_0 \leq G$. We know from [Laz65, Chapitre V, Proposition 2.2.4] that $\Lambda(G_0)$ is a Noetherian ring, and Lemma 3.1 tells us that $\check{V}$ is a finitely generated $\Lambda(G_0)$-module. Thus, the $\Lambda(G_0)$-submodule

$$S^0(V) = \varinjlim_{N} \check{V}^N \leq \check{V}$$

(where N runs over the open normal subgroups of G_0) is automatically also finitely generated; it is then an immediate consequence that there exists some such $N \leq G_0$ satisfying $S^0(V) = \check{V}^N$. Now $S^0(V)$ is a finitely generated $\Lambda(G_0)$-module on which N acts trivially, hence is a finitely generated $E[G_0/N]$-module, implying that it is finite dimensional over E.

In particular, if $V \in \mathrm{Rep}_E^\infty(G)^a$ is irreducible and infinite dimensional over E, then the finite-dimensional $\Lambda(G)$-submodule $S^0(V)$ of the simple $\Lambda(G)$-module $\check{V}$ must be zero. □

Thus, in order to gain useful duality information in natural characteristic, we will need to use higher smooth duals, as defined by Kohlhaase on [Koh17, p. 18-19].

Definition 3.3. For $M \in \mathrm{Mod}_{\Lambda(G)}$, and for $i \geq 0$, we define

$$\Sigma^i(M) = \Sigma^i_G(M) := \varinjlim_{N} \mathrm{Ext}^i_{\Lambda(H)}(E, M),$$

where the limit is taken over the directed family of all open subgroups $H \leq G$.

A priori, these are merely E-vector spaces, however we can view them as representations of G in the following way. Given an open subgroup $H \leq G$ and any $g \in G$, conjugation by g induces an isomorphism $c(g) : \Lambda(H) \to \Lambda(gHg^{-1})$ of E-subalgebras of $\Lambda(G)$. Via this isomorphism, we get an evident equivalence of abelian categories which we denote by

$$c(g)_* : \mathrm{Mod}_{\Lambda(H)} \to \mathrm{Mod}_{\Lambda(gHg^{-1})}\,.$$

Given any $M \in \mathrm{Mod}_{\Lambda(G)}$, the map

$$(m \mapsto gm) : c(g)_*(M) \to M$$

is an isomorphism of $\Lambda(gHg^{-1})$-modules; since $c(g)_*(E) = E$ over $\Lambda(gHg^{-1})$, we conclude that there are E-linear isomorphisms

$$\mathrm{Ext}^i_{\Lambda(H)}(E, M) \cong \mathrm{Ext}^i_{\Lambda(gHg^{-1})}(c(g)_*(E), c(g)_*(M)) \cong \mathrm{Ext}^i_{\Lambda(gHg^{-1})}(E, M).$$

Moreover, one can easily check that these isomorphisms are trivial if $g \in H$, and are compatible with one another as H varies. Thus, we obtain a smooth action of G on the E-vector space $\Sigma^i_G(M)$, so we may view our definition as yielding functors

$$\Sigma^i = \Sigma^i_G : \mathrm{Mod}_{\Lambda(G)} \to \mathrm{Rep}^\infty_E(G).$$

We also note that if $i = 0$, then one can immediately see from

$$\mathrm{Ext}^0_{\Lambda(H)}(E, M) = \mathrm{Hom}_{\Lambda(H)}(E, M) = M^H$$

that the Σ^0 we defined here coincides with that of Definition 2.9.

Definition 3.4. For $i \geq 0$, the *i-th smooth duality functor* $S^i = S^i_G : \mathrm{Rep}^\infty_E(G) \to \mathrm{Rep}^\infty_E(G)$ is defined by

$$S^i = \Sigma^i \circ \check{(\cdot)}.$$

One can easily prove that $(S^i_G)_{i\geq 0}$ form a δ-functor $\mathrm{Rep}^\infty_E(G) \to \mathrm{Rep}^\infty_E(G)$.

4 Key example: supersingular representations of $\mathrm{GL}_2(\mathbb{Q}_p)$

We briefly review the definition of supercuspidal representations. For now, let $F/\mathbb{Q}_p$ be a finite extension, and let $\mathbb{G}$ be a connected reductive F-group; write $G = \mathbb{G}(F)$. We remain in natural characteristic, i.e., we still assume $\mathrm{char}\, E = p$. For $\mathbb{P} = \mathbb{M}\mathbb{N} \leq \mathbb{G}$ a parabolic F-subgroup with provided Levi decomposition, we write P, M, N for the F-points of $\mathbb{P}, \mathbb{M}, \mathbb{N}$, respectively.

Definition 4.1. An irreducible admissible smooth G-representation V over E is *supercuspidal* if it is not isomorphic to any subquotient of the parabolic induction $\mathrm{Ind}^G_P(\sigma)$, where $\mathbb{P}$ is any proper parabolic F-subgroup of $\mathbb{G}$, and σ is an admissible smooth M-representation over E.

We have by [AHHV17, Theorem 5] that an irreducible admissible representation of G is supercuspidal if and only if it is supersingular. In the case we are interested in, there is a full classification of these due to Barthel–Livné in [BL95] and Breuil in [Bre03], summarized briefly in [Koh17, pp. 38-42].

Suppose now that $F = \mathbb{Q}_p$ and $\mathbb{G} = \mathrm{GL}_2$, so $G = \mathrm{GL}_2(\mathbb{Q}_p)$.

Notation 4.2. Let $Z \cong \mathbb{Q}_p^\times$ denote the center of G, and set $G_0 := Z \cdot \mathrm{GL}_2(\mathbb{Z}_p)$. Let V denote a finite dimensional irreducible representation of $\mathrm{GL}_2(\mathbb{F}_p)$ over E, i.e., a Serre weight, which can be inflated to a smooth representation of G_0 via

$$\mathrm{GL}_2(\mathbb{Z}_p) \twoheadrightarrow \mathrm{GL}_2(\mathbb{F}_p)$$

and letting $p \in Z$ act trivially. We write

$$w := \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \alpha := \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix}, \qquad \text{and} \qquad G_n := G_0 \cap \alpha^n G_0 \alpha^{-n}$$

for any $n \in \mathbb{Z}$. Let $K_0 := \mathrm{GL}_2(\mathbb{Z}_p)$ and, for $m \geq 1$, say

$$K_m := \ker\left(\mathrm{GL}_2(\mathbb{Z}_p) \twoheadrightarrow \mathrm{GL}_2(\mathbb{Z}_p/p^m\mathbb{Z}_p)\right).$$

As usual, let $T \leq G$ denote the subgroup of diagonal matrices. Write N and $\overline{N}$ for the subgroups of upper and lower triangular matrices, respectively. We then also set for any $n \geq 1$

$$T_0 := T \cap G_0, \qquad N_n := \alpha^n (N \cap G_0)\alpha^{-n}, \qquad \text{and} \qquad \overline{N}_n = \alpha^{-n}(\overline{N} \cap G_0)\alpha^n.$$

For any such n, there is an Iwahori decomposition $G_n = \overline{N}_0 T_0 N_n$.

We briefly recall the construction of the Hecke operator $T = T_V$ on $\mathrm{ind}_{G_0}^G(V)$. The composition of the natural maps $V^{N_0} \hookrightarrow V \twoheadrightarrow V_{\overline{N}_0}$ is an isomorphism, and hence induces an E-linear map

$$U = U_V : V \to V_{\overline{N}_0} \cong V^{N_0} \hookrightarrow V.$$

Then for $[g, v] \in \mathrm{ind}_{G_0}^G(V)$ denoting the function with support $G_0 g^{-1}$ and value v at g^{-1}, we have that

$$T([g, v]) = \sum_{G_0\alpha G_0 = \bigsqcup x\alpha G_0} [gx\alpha, U(x^{-1}v)].$$

This can also be written as

$$T([g, v]) = [gw\alpha, U(wv)] + \sum_{n \in N_0/N_1} [gn\alpha, U(n^{-1}v)]$$

using the decomposition $G_0\alpha G_0 = w\alpha G_0 \sqcup N_0\alpha G_0$. (We note that Kohlhaase uses an alternate yet isomorphic standardization of induction, but the formula written above is correct in either setting.)

Now the G-representation

$$\pi_V := \mathrm{ind}_{G_0}^G(V)/T\left(\mathrm{ind}_{G_0}^G(V)\right)$$

is irreducible and supersingular, and every supersingular smooth representation of $G = \mathrm{GL}_2(\mathbb{Q}_p)$ over E with trivial action of $p \in Z \subseteq G_0$ is of this form; these claims are proven in [BL95, Proposition 4, Theorem 34] and [Bre03, Théorème 1.1].

Before stating the result of Kohlhaase we are most interested in, we note that $\mathrm{ind}_{G_0}^G(V)$, and hence also π_V, admit central characters which are equal to the central character of V. We will write $\delta_V : G \to E^\times$ for the composition of this central character with the determinant map $\det : G \to \mathbb{Q}_p^\times \cong Z$.

Theorem 4.3. *([Koh17, Theorem 5.13]) If $G = \mathrm{GL}_2(\mathbb{Q}_p)$ and if V is an irreducible representation of G_0 as constructed above, then the supersingular G-representation π_V satisfies*

$$S^i_G(\pi_V) \cong \begin{cases} \pi_{\check{V}} \cong \pi_V \otimes_E \delta_V^{-1} & \text{if } i = 1, \\ 0 & \text{if } i \neq 1, \end{cases}$$

where these are isomorphisms as smooth G-representations over E.

Although the proof of this result is too involved to present in full, we will sketch its main steps. First, we have the following result, also by Kohlhaase (which applies to slightly more general situations, i.e., where the original representation of $\mathrm{GL}_2(\mathbb{F}_p)$ was only assumed to be finite dimensional, not necessarily irreducible):

Proposition 4.4. *([Koh17, Theorem 5.11]) If $G = \mathrm{GL}_2(\mathbb{Q}_p)$ and if V is as above, then $S^i_G(\mathrm{ind}^G_{G_0}(V)) = 0$ for $i \geq 2$.*

We will fully omit the proof of this proposition, as it is quite technical and requires background beyond the scope of this paper. Kohlhaase then manually proves the following claims, where we recall that $S^i_G : \mathrm{Rep}^\infty_E(G) \to \mathrm{Rep}^\infty_E(G)$ is a functor (and hence sends morphisms to morphisms):

Lemma 4.5. *([Koh17, pp. 42-47])*

1. *$S^0_G(T_V)$ is injective.*
2. *$S^1_G(T_V)$ is bijective.*
3. *The restriction of $S^0_G(T_V)$ to $\mathrm{ind}^G_{G_0}(\check{V}) \subseteq S^0_G(\mathrm{ind}^G_{G_0}(V))$ coincides with $T_{\check{V}}$. Thus, we get a G-equivariant map*

$$\pi_{\check{V}} = \mathrm{coker}(T_{\check{V}}) \to \mathrm{coker}(S^0_G(T_V)),$$

which is in fact a bijection.

Now, we apply the δ-functor $(S^i_G)_{i\geq 0}$ to the short exact sequence

$$0 \to \mathrm{ind}^G_{G_0}(V) \xrightarrow{T_V} \mathrm{ind}^G_{G_0}(V) \to \pi_V \to 0$$

in $\mathrm{Rep}^\infty_E(G)$, where T_V is injective by [BL95, Theorem 19], yielding a long exact sequence:

$$\begin{array}{l} 0 \longrightarrow S^0_G(\pi_V) \longrightarrow S^0_G(\mathrm{ind}^G_{G_0}(V)) \xrightarrow{S^0_G(T_V)} S^0_G(\mathrm{ind}^G_{G_0}(V)) \longrightarrow \\ \longrightarrow S^1_G(\pi_V) \longrightarrow S^1_G(\mathrm{ind}^G_{G_0}(V)) \xrightarrow{S^1_G(T_V)} S^1_G(\mathrm{ind}^G_{G_0}(V)) \longrightarrow \\ \longrightarrow S^2_G(\pi_V) \longrightarrow S^2_G(\mathrm{ind}^G_{G_0}(V)) \xrightarrow{S^2_G(T_V)} S^2_G(\mathrm{ind}^G_{G_0}(V)) \longrightarrow \cdots \end{array}$$

We now invoke each of the claims provided in Proposition 4.4 and Lemma 4.5. Since $S^i_G(\mathrm{ind}_{G_0}^G(V)) = 0$ for $i \geq 2$, and $S^1_G(T_V)$ is a bijection, we immediately conclude that $S^i_G(\pi_V) = 0$ for $i \geq 2$. What remains is the following exact sequence:

$$0 \to S^0_G(\pi_V) \to S^0_G(\mathrm{ind}_{G_0}^G(V)) \xrightarrow{S^0_G(T_V)} S^0_G(\mathrm{ind}_{G_0}^G(V)) \to S^1_G(\pi_V) \to 0.$$

Because also $S^0_G(T_V)$ is injective, we conclude that $S^0_G(\pi_V) = 0$, which yields the following short exact sequence:

$$0 \to S^0_G(\mathrm{ind}_{G_0}^G(V)) \xrightarrow{S^0_G(T_V)} S^0_G(\mathrm{ind}_{G_0}^G(V)) \to S^1_G(\pi_V) \to 0.$$

Thus, $S^1_G(\pi_V) \cong \mathrm{coker}(S^0_G(T_V)) \cong \pi_{\check{V}}$, where the last isomorphisms follows from Lemma 4.5.

All that remains is to establish the final formula $\pi_{\check{V}} \cong \pi_V \otimes_E \delta_V^{-1}$. As in [Koh17], one can prove this using the following steps.

First, show $\check{V} \cong V \otimes_E$ are isomorphic representations of G. Use this to conclude that, under this isomorphism, the operators $U_{\check{V}}$ and $U_V \otimes_E 1$ correspond to one another. Use this to then conclude that, under the G-equivariant isomorphisms

$$\mathrm{ind}_{G_0}^G(\check{V}) \cong \mathrm{ind}_{G_0}^G(V \otimes_E \delta_V^{-1}) \cong \mathrm{ind}_{G_0}^G(V) \otimes_E \delta_V^{-1},$$

the operator $T_{\check{V}}$ corresponds to $T_V \otimes_E 1$. Thus,

$$\pi_{\check{V}} \cong \pi_V \otimes_E \delta_V^{-1},$$

completing the sketch of the proof of Theorem 4.3.

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